Project Proposal: Comparative Analysis of Machine Learning Models for Highly Imbalanced Credit Card Fraud Detection

AI COURSE - MACHINE LEARNING TRACK

1. Project Objective

The primary goal of this project is to build and evaluate multiple machine learning models to classify credit card transactions as either legitimate or fraudulent. Because fraudulent transactions make up less than 0.2% of all data, the project will specifically focus on addressing extreme class imbalance.

The analysis will demonstrate why standard metrics like "Accuracy" are deeply flawed for this domain, and will instead evaluate models based on their ability to minimize False Negatives (maximizing Recall) without triggering excessive False Positives (maintaining Precision).

2. Dataset Description & Feature List

> **Source:** Kaggle (Credit Card Fraud Detection Dataset)
> **Size:** 284,807 transactions, containing 492 frauds.

Due to confidentiality issues, the dataset does not provide the original background features of the transactions (like customer name or location). Instead, it provides features that have undergone a Principal Component Analysis (PCA) transformation.

Full Feature List:

- **Time:** Number of seconds elapsed between this transaction and the first transaction in the dataset.
- **Amount:** The monetary value of the transaction. (Highly variable; will require scaling).
- **V1 through V28:** 28 anonymized, numerical features resulting from the PCA transformation.
- **Class (Target Variable):** `0` represents a legitimate transaction, and `1` represents a fraudulent transaction.

3. Methodology Overview

The project will follow a standard Data Science pipeline, breaking down the work into four distinct phases:

Phase 1: Exploratory Data Analysis (EDA)

- Verify the severe class imbalance (99.8% vs 0.2%).
- Analyze the distribution of the *Amount* and *Time* features for both classes (e.g., do fraudulent transactions tend to be larger amounts?).
- Check for missing or null values (fortunately, this dataset usually has none).

Phase 2: Data Preprocessing

- **Feature Scaling:** Since features V1-V28 are already PCA-transformed (and scaled), we must scale *Time* and *Amount* using a `StandardScaler` or `RobustScaler` so they do not dominate the models.
- **Data Splitting:** Split the data into an 80% training set and a 20% testing set using a stratified split (to ensure the 0.2% fraud ratio is maintained in both sets).
- **Addressing the Imbalance:** Apply a technique to handle the skewed data on the training set. Options include:
  - *Algorithmic adjustment:* Using `class_weight='balanced'` inside the models.
  - *Data adjustment:* Using SMOTE (Synthetic Minority Over-sampling Technique) to generate synthetic fraud examples for the models to learn from.

Phase 3: Model Implementation

Four distinct classification models will be trained on the preprocessed data:

- **Logistic Regression:** A linear baseline model. Fast and interpretable.
- **Decision Tree:** A non-linear model that creates interpretable "if/then" rules.
- **Random Forest:** An ensemble model that builds multiple decision trees to reduce overfitting and improve variance.
- **Support Vector Machine (SVM):** A model that finds the optimal hyperplane separating the two classes (will likely require a subset of data due to high computational time on 280k+ rows).

Phase 4: Evaluation & Comparative Analysis

Each model will be evaluated on the unseen 20% testing data. The analysis will compare:

- **Confusion Matrices:** Visually showing True Positives, True Negatives, False Positives, and False Negatives.
- **Metrics Comparison:** A side-by-side table of Accuracy, Precision, Recall, and F1-score.
- **Result Interpretation:** A detailed written conclusion focusing on the trade-off between Precision and Recall. The final recommendation will identify which model is best suited for real-world deployment by prioritizing the F1-score and Recall of the minority class.

4. Expected Deliverables

- A Jupyter Notebook (`.ipynb`) containing all Python code, heavily commented.
- Visualizations of the Confusion Matrices for all 4 models.
- A final written report (or presentation) analyzing the results and fulfilling the comparative analysis requirement of the course.